Unit: 6 Unit 6: Systems of Linear Equations and Inequalities
Topic: Solving Systems by Graphing

Name: _____
Date: _____

Advanced (Habanero)

Q1. What is the solution to the system $x+y=2$ and $x-y=-2$?
(A) $(-2,4)$
(B) $(2,0)$
(C) $(0,-2)$
(D) $(1,1)$
(E) $(4,-2)$

Q2. Which of the following is a step to graph a system of linear equations?
(A) Find the y-intercept of the second equation
(B) Graph the first equation and find the x-intercept
(C) Solve the system using substitution
(D) Graph both equations on the same coordinate plane
(E) Find the slope of the first equation

Q3. The system $x+2y=4$ and $3x-2y=5$ has what kind of solution?
(A) One solution
(B) No solution
(C) Infinitely many solutions
(D) Two solutions
(E) Three solutions

Q4. What is the point of intersection between the lines $y=2x-1$ and $y=-3x+4$?
(A) $left($
rac55,
rac35 ight)
(A) $left(-$
rac55,
rac35 ight)
(A) $left($
rac55, 1 ight)
(A) $left(1, 1ight)$
(B) $left($
rac55,
rac95 ight)

Q5. The system of inequalities $ygeqx+1$ and $yleq-x-2$ has what kind of solution set?
(A) No solution
(B) Infinitely many solutions
(C) One solution
(D) A bounded region
(E) An unbounded region

Q6. To find the point of intersection of the lines $y=3x-2$ and $y=2x+1$, we can set the two equations $3x-2=2x+1$ and solve for x.
(A) False
(B) True
(C) Sometimes true
(D) Only for parallel lines
(E) Only for perpendicular lines

Q7. The system $y=x+2$ and $y=x-2$ has what kind of solution set?
(A) Infinitely many solutions
(B) One solution
(C) No solution
(D) Two solutions
(E) Parallel lines

Q8. How many solutions does the system $x+y=2$ and $2x+2y=4$ have?
(A) Infinitely many solutions
(B) One solution
(C) No solution
(D) Two solutions
(E) Three solutions

Open Ended Questions (FRQ)

Q9. Graph the system of linear equations $y=2x-1$ and $y=-x+3$. Identify the point of intersection and explain its significance.

Q10. Consider the system of linear inequalities $ygeqx-2$ and $yleq-2x+1$. Graph the solution set and describe the bounded region.

Chef's Tips

- Provide clear instructions to students.
- Ensure students have necessary materials.
- Encourage students to ask questions if they are unsure.